# Evaluation: Gate Criteria, QA Probes, and Failure Discovery

## Evaluation: Gate Criteria, QA Probes, and Failure Discovery I

The evaluation section is configured to name readiness criteria, connect criteria to artifacts, separate local checks from publication readiness, and make failure probes visible. The local readiness surface is not a human preference score; it is a set of deterministic checks that connect generated manuscript claims to source files and pipeline gates.

The active criteria use analysis, copy, pytest, validation as gate labels and inspect artifacts such as token-injection flow, quality-gate matrix, configured-field figures. A passing run means the exemplar is locally render-ready: placeholders resolve, token provenance is present, figure references are registered, evidence scanning has not found unsupported numbers, and project design overlays remain internally consistent.

That readiness is deliberately narrower than publication readiness. A local pass does not imply reader-preference results or empirical validation. It means the tracked project tree can regenerate the committed artifact surface through its declared pipeline.

The QA probes are Method row completeness, Field-origin visibility, Placeholder survival, Provenance completeness, Section-switch observability, Figure registry coverage, Method-figure alignment, Evidence cleanliness, Fork readiness, Copied-output parity, Digest invariant review, Claim-ledger alignment, Review packet completeness, Fork migration sufficiency. They are phrased as questions so they can be reused by reviewers and by forks of the exemplar: did the placeholder disappear, did the provenance survive, did the figure registry cover every reference, and did copied outputs preserve the same evidence surface that validation inspected?

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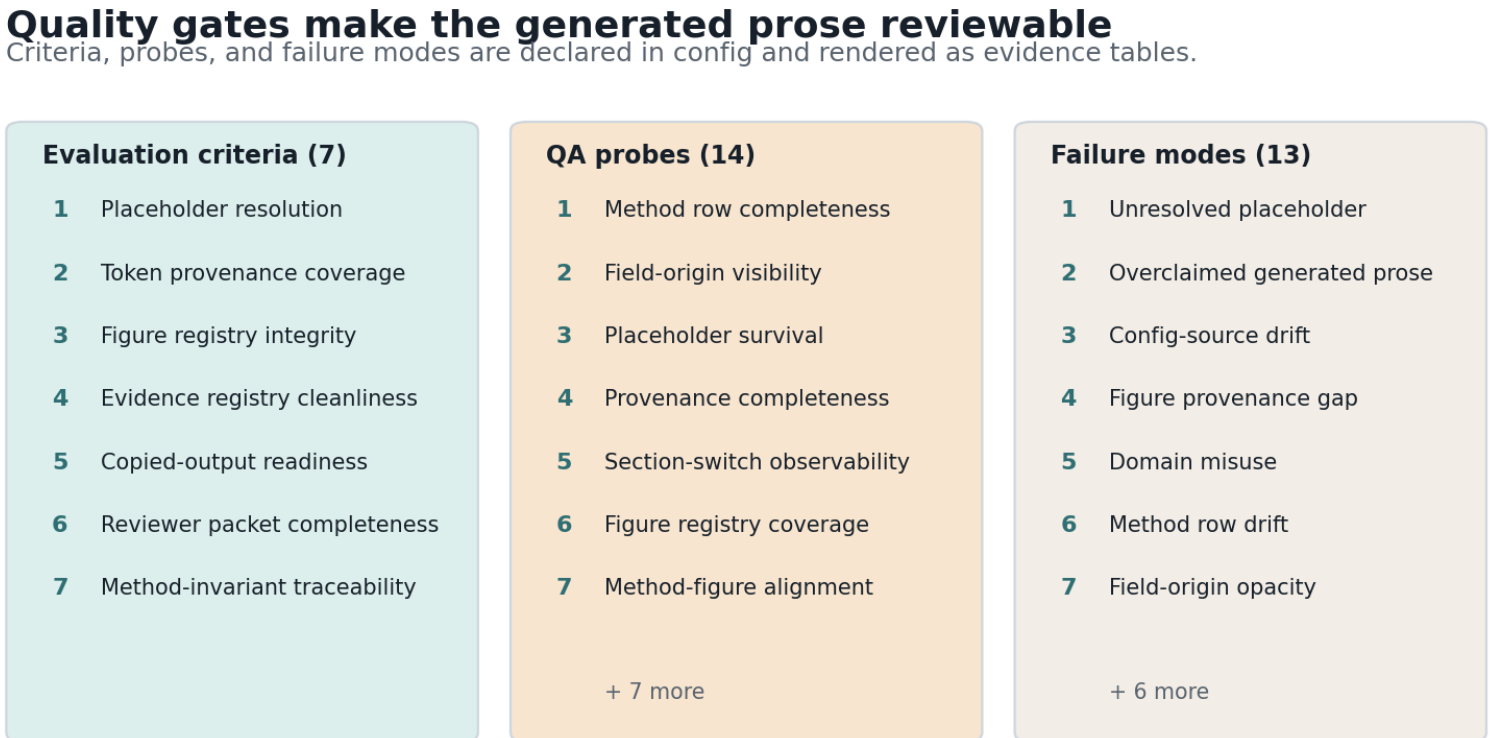



Figure 1: Quality gate matrix

# Evaluation Criteria I

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| Criterion | Target | Evidence | Gate |
|---|---|---|---|
| Placeholder resolution | No unresolved uppercase manuscript placeholders remain in output/manuscript or rendered web output. | tests/test_manuscript_variables.py and rg unresolved-token scan. | `pytest` |
| Token provenance coverage | Every selected token maps to category, selected value, section, and config pointer. | output/reports/injection_trace.json and output/data/token_inventory.json. | `analysis` |
| Figure registry integrity | Every referenced figure label is present in output/figures/figure_registry.json. | scripts/pipeline/stage_04_validate.py figure registry check. | `validation` |
| Evidence registry cleanliness | Generated manuscript numbers and claims pass the project evidence registry. | output/reports/evidence_registry.json | `validation` |
| Copied-output readiness | PDF, HTML, slides, figures, data, and reports copy into output/templates/template_madlib. | scripts/pipeline/stage_05_copy.py output statistics. | `copy` |
| Reviewer packet completeness | Hydrated Markdown, rendered PDF, web output, slides, figures, data, reports, validation results, and copy statistics are all present for review. | Stage 04 validation report and Stage 05 output_statistics.json. | `copy` |
| Method-invariant traceability | Token choices can be explained only by seed, slot, category, ordinal, and ordered category inventory. | tests/test_tokens.py and generated Methods digest prose. | `pytest` |

# Quality Probes I

# Quality Probes II

| Probe | Question | Passing signal | Artifact |
|---|---|---|---|
| Method row completeness | Does the protocol table cover schema intake, token planning, composition, figures, validation, copy, and review handoff? | method_protocol includes rows for every major pipeline responsibility. | `manuscript/config.yaml and output/data/section_plan.json` |
| Field-origin visibility | Can a reviewer tell which visible fields were authored and which were defaulted? | Configured-field inventory and summary tables report explicit and defaulted paths. | `output/data/configured_field_inventory.json` |
| Placeholder survival | Did any source token survive hydration? | No uppercase placeholders are found in generated manuscript or web files. | `output/manuscript and output/web` |
| Provenance completeness | Can every selected token be traced to a category, section, value, and config key? | The injection trace and token inventory contain one row for each generated token. | `output/reports/injection_trace.json` |
| Section-switch observability | Does a disabled section resolve to a visible explanation? | Disabled section bodies cite their controlling section condition. | `output/data/section_plan.json` |
| Figure registry coverage | Does every manuscript figure reference have a generated registry entry? | Figure registry validation passes. | `output/figures/figure_registry.json` |
| Method-figure alignment | Do method figures describe generated data rather than decorative or unsupported claims? | Figure registry captions, nonblank PNG tests, and manual visual QA align with artifact data. | `output/figures` |
| Evidence cleanliness | Do generated claims stay within local evidence boundaries? | Evidence registry validation passes without unsupported claims. | `output/reports/evidence_registry.json` |
| Fork readiness | Does the authoring contract tell downstream forks what to extend before adding domain claims? | Authoring obligations cite config diffs, claim ledger updates, validators, and full reruns. | `output/manuscript/10_authoring_contract.md` |
| Copied-output parity | Did copied deliverables preserve the validated project output surface? | Copy-stage statistics include PDF, HTML, slides, figures, data, and reports. | `output/templates/template_madlib` |
| Digest invariant review | Are the allowed token-selection | Methods prose names the digest | `src/template_madlib/tokens.p` |